INCH-POUND

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# COMMERCIAL ITEM DESCRIPTION

## COMPASSES, MAGNETIC: SURVEYOR'S AND TRANSIT, POCKET; WITH OPTIONAL BALL AND SOCKET JOINT OR BALL AND SOCKET HEAD, AND JACOB'S STAFF

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GG-C-560D.

This description covers two types of magnetic compasses, and includes the optional ball and socket joint, or ball and socket head, and Jacob's staff. Type I compasses are intended for use in topographical surveying field work done by the U.S. Forest Service. Type II compasses are used in geology and survey reconnaissance.

General salient characteristics.

Design: The compasses shall consist of a compass box, clinometer, magnetic needle, circular level vial, mechanism to permit declination adjustment, front and rear sights, graduated circles and scales, and when specified herein, a Jacob's staff, ball and socket joint, or ball and socket head and a carrying case.

Classification. Magnetic compasses covered by this specification shall be the following types, classes, and styles as specified:

Type I - Surveyor's magnetic compass using ball and socket joint.
Type II - Magnetic compass (pocket transit) using ball and socket head.

Beneficial comments recommendations, additions, deletions clarifications, etc. and other data which may improve this document should be sent by letter to: U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA.

AMSC N/A FSC 6675

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Class 1 - 0 to 90 degree quadrant dial.
Class 2 - 0 to 360 degree circle dial.

Style A - With ball and socket joint, or ball and socket head, Jacob's staff.
Style B - Without ball and socket joint, or ball and socket head, and without Jacob's staff.

Dial Marking. Class 1 and class 2 compass dials shall be graduated in 1 degree increments. Every increment shall be marked with a tick mark. Every tenth degree shall be numbered (i.e., 10, 20, etc.).

Class 1. Class 1 compass dials shall be divided into 90 degree quadrants. The zero degree graduation shall be in the North and South directions. The 90 degree graduation shall be in the East and West directions.

Class 2. Class 2 compass dials shall be graduated in a complete circle. The zero degree graduation shall be in the North direction, the 90 degree graduation shall be in the East direction, and the 180 degree graduation shall be in the South direction.

Material:

Metal. The metal, except that of the needle, shall be nonferrous and corrosion resistant. Needles shall be plated or otherwise durably coated to render them corrosion resistant.

Compass box. The compass box shall be a non-magnetic material and shall have a transparent window affording full visibility of the azimuth scale, the clinometer level vial scale and index, and the compass level vial. The compass box shall have an external screw driver-type adjustment so that the azimuth scale can be rotated with respect to the index.

Magnetic needle. The magnetic needle shall have magnets permanently attached or shall be magnetized to be sensitive to the earth's magnetic field. The needle shall be designed and constructed to assure long retention of magnetism and shall indicate magnetic North when the box is rotated 360 degrees. The needle shall be freely supported on a pivot of nonmagnetic hardened material that provides minimum friction. The needle shall be designed in such a manner as to permit use or be provided with a balance weight which can be moved readily to any position and on either end to assure correct balance in any latitude between 85 degrees North magnetic latitude and 65 degrees South magnetic latitude. The needle shall be self-aligning to accurately indicate direction of magnetic North and shall be marked to readily indicate North. The ends of the needle shall be tapered to a fine point or



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marked so that 0.50 degrees can be estimated. There shall be no apparent difference in the reading of the ends of the needle. The needle shall be equipped with a lift mechanism to secure it when not in use. The damping time of the needle shall not exceed 15 seconds when the needle is deflected.

Jacob's staff. The Jacob's staff shall be a hardwood or nonferrous metal staff, not less than 54 inches long, threaded 0.625-11 UNC-2A for a type I compass and 0.25-20 UNC-2A for type II compass to mate with the female end of the ball and socket joint or ball and socket head. The other end of the staff shall be equipped with a pointed metal shoe for penetrating the ground. The staff shall be one piece or collapsible and shall weigh not more than 4 pounds.

Ball and socket joint or ball and socket head. A ball and socket joint or ball and socket head to attach the compass to the Jacob's staff shall be constructed to permit tilting the compass through 90 degrees from the horizontal to the vertical plane and also permit full rotation of the compass in azimuth through 360 degrees. The ball joint shall be equipped with a knurled collar to tighten the joint in tilted position and a knurled thumbscrew to tighten the joint in the azimuth position. The male end of the ball joint shall be threaded 0.50-20 UNF-2A to mate with that in the compass base female recessed fitting, which shall be at least 0.250-inch deep for a type I compass and the type II compass shall be furnished with a bracket to attach the compass to the ball joint. The other end of the ball joint shall be threaded 0.625-11 UNC-2B for a type I compass and 0.250-20 UNC-2B for type II compass to fit the Jacob's staff.

Carrying case. When required, a sewed leather case with a sling strap shall be provided to hold the compass. When applicable, the carrying case shall be of case leather, mildew resistant treated, grain side smooth with the sides of the case formed. A hold down strap and a nonferrous metal fastening stud shall be provided for the closing flap.

Tolerances. The tolerances for the beveled knurled revolving ring, compass dial, and raised compass ring shall be ±10 minutes per degree graduation, with no more than ±30 minutes per quadrant. The error shall be noncumulative over the entire circle.

Specific salient characteristics.

Type I, surveyor's magnetic compass using ball and socket joint.

Beveled knurled revolving ring. The beveled, knurled revolving ring shall be aluminum and shall be graduated in single degrees both ways from zero in four 90-degree quadrants, with a sighting mark at each quadrant.



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Compass dial. The compass dial shall be marked to indicate the four cardinal compass points. East and west points shall be transposed 180 degrees from normal map positions. A clinometer scale for reading vertical angles shall be graduated in 1-degree increments for 180 degrees, and numbered every 10 degrees in both directions from zero to 90 degrees. The zero point shall coincide with the west cardinal point on the compass dial.

Raised compass ring. The raised compass ring shall be graduated in single degrees and numbered every 10 degrees in both directions from zero to four 90-degree quadrants. A mechanism shall be provided to adjust the azimuth of the raised compass reading for declination. The mechanism shall be enclosed within the compass box except for the operating pinion, which shall extend through to the outside of the compass box terminating in a slotted screwhead to facilitate the adjustment from outside the compass box. The mechanism shall be nonmagnetic, corrosion-resistant metal and shall be constructed in such a manner that dust or moisture cannot enter the compass box where the adjustment screw terminated. A fixed index pointer shall be permanently secured to the inside wall of the compass box and shall extend over the raised compass ring to serve as a reference point for adjusting the raised compass ring. The index pointer shall coincide with the north point of the compass dial.

Pendulum clinometer. A clinometer pendulum shall be enclosed in the compass box for reading vertical angles when the compass is placed vertically on its edge on a level surface. The pendulum shall swing freely from an axis concentric with the pivot of the compass needle. The vertical angle shall be indicated by the position of the pendulum against the clinometer scale on the compass dial.

Spirit level. A circular hermetically sealed spirit level for leveling the compass shall be located and secured to the inside of the compass box. The level shall not cover any essential inscriptions.

Compass box. The compass box shall not be more than 4 inches in diameter.

Magnetic needle. The magnetic needle shall be not less than 3 inches long and shall be equipped with a stop and a jewel setting.

Folding sights. Two folding sights shall be secured to the compass box by not less than two screws for each sight. The sights shall be hard forged brass or bronze and shall be constructed to permit their being raised to a vertical position for sighting objects or to be folded down flat over the compass glass when not in use. When in the vertical sighting position, the sights shall project



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above the revolving ring on top of the compass box to a distance equal to the diameter of the revolving ring. The center of the sights shall be positioned in the north-south axis of the compass dial and also to the centerline of the base parallel to two of the beveled edges of the base. The rear (south) folding sight shall have a narrow vertical peep slot centered in the sight. The length of the slot shall be 75 percent of the total length of the sight. The rear sight shall be located on the south end of the compass box and in line with the north-south axis. The front (north) sight shall have a wider slot, 50 percent of the width of the sight, equipped with a vertical sighting line from 0.010 to 0.015 inch wide running the length of the slot. The line and slot shall be centered on the north-south axis of the compass dial at the north end of the compass box.

Compass base. A 4.50 by 4.50-inch aluminum, nickel, silver, or other non-magnetic material base shall be secured to the compass box by not less than four countersunk screws with the compass box positioned in the center of the base. The relative position of the base and compass box, in azimuth, shall be that the flat-sides of the base shall be parallel to the north-south and east-west axes of the compass dial. The base shall be smooth and flat on the bottom to permit close and accurate positioning on a flat surface. The base shall have cut away or recessed space for screw adjustment devices and fittings. An internal fitting threaded 0.50-20 UNF-2B, shall be provided in the base to accommodate the male end of the ball and socket join to ball and socket head. The fitting shall not permit dust or moisture to enter the compass box. The base shall have four beveled and graduated edges. Two adjacent edges shall be graduated in 0.50 degree increments, ±30 seconds, through 90 degrees and numbered every 10 degrees up to 80 degrees in both directions. The 45-degree mark shall be at the intersecting corner of the two adjacent edges. Another edge shall be graduated in 0.125-inch increments, ±0.0156-inch and numbered every 1 inch. The other edge shall be graduated in 0.10-inch increments, ±0.01-inch, and numbered every 1 inch.

Specific salient characteristics:

Type II, magnetic compass (pocket transit) using ball and socket head.

Compass dial. The compass dial shall be graduated in 1-degree increments and numbered every graduated 10 degrees.

Declination of adjustment. A mechanism shall be provided to adjust the position of the azimuth of the compass dial not less than 50 degrees in either direction.



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Index pointer. An index pointer shall be permanently secured to the inside wall of the compass box and shall extend over the compass dial. The index pointer shall be in line with the front sight and shall coincide with the north point of the compass.

Clinometer. The clinometer shall consist of a tubular vial, a clinometer scale and a vernier index mounted on the compass box in such a manner that the clinometer may be rotated by an external leveler. The clinometer shall have its zero at the West cardinal point. Graduations shall extend on both sides of the zero and shall be in degree and percent declination. The degree scale shall be graduated every 1 degree, ±.50 degree, shall extend 90 degrees on each side of zero, and shall be numbered every 10 degrees. The vernier shall permit reading to 5 minutes. The percent scale shall extend 100 percent in both directions. Graduations shall be by 5 percent increments. Each 20 percent graduation shall be numbered.

Spirit level. A circular hermetically sealed spirit level shall be secured to the inside of the compass box in the Southeast or Northeast quadrant. The level shall not cover any essential inscriptions.

Dimensions and weights. The closed compass shall be not more than 4 by 4 inches, with rounded corners, by not more than 1.375 inches thick. The compass with carrying case shall weigh not more than 10 ounces.

Magnetic needle. The magnetic needle shall be between 2.000 and 2.250 inches long, with jeweled center and a stop.

Mirror and cover. A mirror shall be bonded or otherwise mounted on a hinged cover attached to the south end of the compass box. The cover and the mirror shall provide a 0.375 by 0.4687-inch clear glass window, an engraved center-line which will serve as an alternate front sight, and a folding front sight tip. Hinge friction shall be such as to permit setting of the mirror at any angle between full-closed and full-open position of the cover. Hinge stops shall be provided at 180 degrees open position. In the closed position, the cover shall actuate a lifting mechanism in the compass box to lift the magnetic needle from its pivot and hold the needle in place against the compass box window.

Cover latch. The cover shall latch into the closed position by use of a spring detent or closely controlled mechanical interference. The cover shall open by use of fingergrips (either natural or machined in) or by use of a push-to-open plunger-type spring release. When no mechanical opening aid is provided, fingergrips will be provided on the exterior of the cover and compass box.



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North sight. The sight attached to the north end of the compass box shall fold through 180 degrees +10 degrees with a tip that folds 60 degrees ±7 degrees upward from open position. The tip shall be 0.375-inch long ±0.03125-inch. The rear sight shall be 2.75-inch long ±0.0625-inch, including the folding tip.

Workmanship. The magnetic compasses shall have a uniform unmarred finish. Each component shall be constructed to facilitate an accurate fit of mating parts. Internal and external parts shall be free from residue of fabrication and shall be free of any defects which affect serviceability or appearance.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Quality assurance: The magnetic compasses shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Design not as specified.
2. Type not as specified.
3. Class not as specified.
4. Style not as specified.
5. Dial marking not as specified.
6. Material not as specified.
7. Compass box not as specified.
8. Magnetic needle not as specified.
9. Jacob's staff not as specified.
10. Ball and socket joint or ball and socket head not as specified.
11. Carrying case not as specified.
12. Tolerances not as specified.
13. Packaging and marking not as specified.



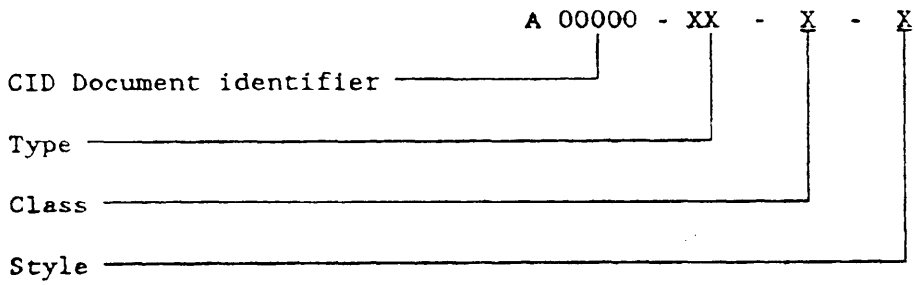
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Regulatory requirements. In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

Part or identifying number (PIN). The part or identifying number for compasses, magnetic, surveyors and transit shall be as follows:




Notes:

1. When the commercial item description is used for acquisition, the commercial item certification clause must appear in the solicitation.

2. Purchaser should specify type, class, and style required and when carrying case is required.

FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.



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MILITARY INTEREST:

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Army - ME
Air Force - 99

Review activities
Air Force - 82
DLA - GS

User:
Navy - YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
USDA-AFS, SCS
INT-BLM

PREPARING ACTIVITY:
Army - ME

Project 6675-0888



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